

Shangtong Zhang

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School of Engineering and Applied Science
University of Virginia
Charlottesville, VA, 22903

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RESEARCH INTEREST

The goal of my research is to solve sequential decision making problems in a scalable and reliable way. Currently, I focus on Reinforcement Learning (RL) as a solution method. In particular, I work on stochastic approximations for RL, theories and algorithms of RL, and applications by RL.

ACADEMIC EMPLOYMENTS

Assistant Professor

Aug 2022 - Now

Department of Computer Science
University of Virginia, VA, United States

Research Scientist Interns

Microsoft Research Montreal
DeepMind London
Microsoft Research Montreal

Jun 2021 - Sep 2021
Feb 2021 - Jun 2021
Jun 2020 - Aug 2020

EDUCATION

Doctor of Philosophy, Computer Science

Oct 2018 - Jul 2022

University of Oxford, Oxford, United Kingdom
Advisor: Prof. Shimon Whiteson
Thesis: Breaking the Deadly Triad in Reinforcement Learning

Master of Science, Computer Science

Sep 2016 - Jul 2018

University of Alberta, Edmonton, Canada
Advisor: Prof. Richard S. Sutton
Thesis: Learning with Artificial Neural Networks

Bachelor of Science, Computing Science

Sep 2012 - Jul 2016

Fudan University, Shanghai, China

HONORS

Google Research Award, 2026
NeurIPS Top Area Chair, 2025
Cisco Faculty Research Award, 2025
NSF CAREER Award, 2025
QuantCo Spotlight (Best Paper) Award at the ICML In-Context Learning Workshop, 2024
Rising Stars in AI by KAUST AI Initiative, 2024
AAAI New Faculty Highlights, 2023

IFAAMAS Victor Lesser Dissertation Award (Runner-Up), 2022
 Alf Weaver Junior Faculty Fellowship, UVA, 2022 - 2027
 ICLR Outstanding Reviewer, 2021
 NeurIPS Reviewer Award, 2020
 ICML Reviewer Award, 2020
 AAMAS Best Paper Award, 2020
 Light Senior Scholarship, St Catherine’s College, University of Oxford, 2020
 EPSRC Studentship, University of Oxford, 2018 - 2022
 EMC Scholarship, Fudan University, 2014

PUBLICATIONS

Advisees of SZ are underlined; * indicates equal contribution; [†] indicates equal advising.

Preprints

9. *Inductive Feedback for Mixed-Policy Distillation.*
Amir Moeini, Huaijiang Zhu, Daniel Havir, **Shangdong Zhang**.
 arXiv Preprint 2026.
8. *OR-Transformer: Scaling Real-Time Decision-Making to 1,000 Items.*
 Shuze Liu, David Simchi-Levi, Claire Chen, Chutong Gao, **Shangdong Zhang**.
 arXiv Preprint 2026.
7. *Robust Data-Collection Policy Learning for Low-Variance Online Policy Evaluation.*
 Claire Chen, Shuze Liu, Licheng Luo, Rohan Chandra, Nan Jiang, **Shangdong Zhang**.
 arXiv Preprint 2026.
6. *Convergence and Emergence of In-Context Reinforcement Learning with Chain of Thought.*
Zixuan Xie, Xinyu Liu, Rohan Chandra, **Shangdong Zhang**.
 arXiv Preprint 2026.
5. *Beyond Linear Attention: Softmax Transformers Implement In-Context Reinforcement Learning.*
Zixuan Xie, Xinyu Liu, Claire Chen, Shuze Liu, Rohan Chandra, **Shangdong Zhang**.
 arXiv Preprint 2026.
4. *Multi-agent DRL-based Lane Change Decision Model for Cooperative Planning in Mixed Traffic.*
 Zeyu Mu, **Shangdong Zhang**, B. Brian Park.
 arXiv Preprint 2026.
3. *Prompt-Driven Domain Adaptation for End-to-End Autonomous Driving via In-Context RL.*
 Aleesha Khurram, Amir Moeini, **Shangdong Zhang**, Rohan Chandra.
 arXiv Preprint 2025.
2. *Extensions of Robbins-Siegmund Theorem with Applications in Reinforcement Learning.*
Xinyu Liu, Zixuan Xie, **Shangdong Zhang**.
 arXiv Preprint 2025.
1. *Almost Sure Convergence Rates and Concentration of Stochastic Approximation and Reinforcement Learning with Markovian Noise.*
 Xiaochi Qian*, Xinyu Liu*, Zixuan Xie*, **Shangdong Zhang**.
 arXiv Preprint, 2024.

Invited Articles

1. *A New Challenge in Policy Evaluation*.
Shangtong Zhang.
AAAI Conference on Artificial Intelligence (**AAAI**), 2023.
New Faculty Highlights Program.

Refereed Journals

5. *Almost Sure Convergence of Linear Temporal Difference Learning with Arbitrary Features*.
Jiuqi Wang, **Shangtong Zhang**.
Journal of Machine Learning Research (**JMLR**), 2026.
4. *The ODE Method for Stochastic Approximation and Reinforcement Learning with Markovian Noise*.
Shuze Liu, Shuhang Chen, **Shangtong Zhang**.
Journal of Machine Learning Research (**JMLR**), 2025.
3. *Global Optimality and Finite Sample Analysis of Softmax Off-Policy Actor Critic under State Distribution Mismatch*.
Shangtong Zhang, Remi Tachet des Combes[†], Romain Laroche[†].
Journal of Machine Learning Research (**JMLR**), 2022.
2. *Truncated Emphatic Temporal Difference Methods for Prediction and Control*.
Shangtong Zhang, Shimon Whiteson.
Journal of Machine Learning Research (**JMLR**), 2022.
1. *MLPack 3: A Fast, Flexible Machine Learning Library*.
Ryan Curtin, Marcus Edel, Mikhail Lozhnikov, Yannis Mentekidis, Sumedh Ghaisas, **Shangtong Zhang**.
Journal of Open Source Software (**JOSS**), 2018.

Refereed Conference Papers

44. *MathlibPR: Pull Request Merge-Readiness Benchmark for Formal Mathematical Libraries*.
Zixuan Xie*, Xinyu Liu*, **Shangtong Zhang**.
Conference on Neural Information Processing Systems (**NeurIPS**), 2026.
Acceptance Rate 25.8% (Evaluations and Datasets Track)
43. *Almost Sure Convergence Rates of Stochastic Approximation and Reinforcement Learning via a Poisson-Moreau Drift*.
Xinyu Liu, Zixuan Xie, **Shangtong Zhang**.
Conference on Neural Information Processing Systems (**NeurIPS**), 2026.
Acceptance Rate 25.7%
42. *On the Divergence of Differential Temporal Difference Learning without Local Clocks*.
David Antrobus, **Shangtong Zhang**.
Conference on Neural Information Processing Systems (**NeurIPS**), 2026.
Acceptance Rate 25.7%
41. *Predicting Plasticity in Deep Continual Learning: A Theoretical Perspective*.
Jiuqi Wang, Jayanth Srinivasa, Claire Chen, Shuze Liu, Ali Payani, **Shangtong Zhang**.
Conference on Neural Information Processing Systems (**NeurIPS**), 2026.
Acceptance Rate 25.7%
40. *Latent Q-Barrier Shielding for Safe In-Context Reinforcement Learning*.
Minjae Kwon*, Amir Moeini*, **Shangtong Zhang**, Lu Feng.
Conference on Neural Information Processing Systems (**NeurIPS**), 2026.
Acceptance Rate 25.7%
39. *InfRL: Inference-time Reinforcement Learning for Research Idea Optimization*.
Sikun Guo, Amir Hassan Shariatmadari, Jiuqi Wang, Albert Huang, Stefan Bekiranov, **Shangtong Zhang**, Aidong Zhang.

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2026.
Acceptance rate: 30.8% (AI4Sciences Track)
38. *Towards Formalizing Reinforcement Learning Theory: A Robbins-Siegmund Approach.*
Shangdong Zhang.
Reinforcement Learning Conference (**RLC**), 2026.
Acceptance rate: 33.9%
 37. *Convergence of Two-Timescale Markovian Stochastic Approximations with Applications in Reinforcement Learning.*
Vagul Mahadevan, Claire Chen, Shuze Liu, **Shangdong Zhang.**
International Conference on Machine Learning (**ICML**), 2026.
Acceptance rate: 26.6%
 36. *Safe In-Context Reinforcement Learning.*
Amir Moeini^{*}, Minjae Kwon^{*}, Alper Kamil Bozkurt, Yuichi Motai, Rohan Chandra, Lu Feng,
Shangdong Zhang.
International Conference on Machine Learning (**ICML**), 2026.
Acceptance rate: 26.6%
 35. *MathlibLemma: Folklore Lemma Generation and Benchmark for Formal Mathematics.*
Xinyu Liu, Zixuan Xie, Amir Moeini, Claire Chen, Shuze Liu, Yu Meng, Aidong Zhang,
Shangdong Zhang.
International Conference on Machine Learning (**ICML**), 2026.
Acceptance rate: 26.6%
 34. *Reward Is Enough: LLMs Are In-Context Reinforcement Learners.*
Kefan Song^{*}, Amir Moeini^{*}, Peng Wang, Lei Gong, Rohan Chandra, **Shangdong Zhang**[†],
Yanjun Qi[†].
International Conference on Learning Representations (**ICLR**), 2026.
Acceptance rate: 28%
 33. *Adaptive Policy Selection and Fine-Tuning under Interaction Budgets for Offline-to-Online Reinforcement Learning.*
Alper Bozkurt, Xiaoan Xu, **Shangdong Zhang**, Miroslav Pajic, Yuichi Motai.
Annual Learning for Dynamics & Control Conference (**L4DC**), 2026.
Acceptance rate: not released
 32. *Almost Sure Convergence of Differential Temporal Difference Learning for Average Reward Markov Decision Processes.*
Ethan Blaser, Jiuqi Wang, **Shangdong Zhang.**
International Conference on Artificial Intelligence and Statistics (**AISTATS**), 2026.
Acceptance rate: 28%
 31. *Asymptotic and Finite Sample Analysis of Nonexpansive Stochastic Approximations with Markovian Noise.*
Ethan Blaser, **Shangdong Zhang.**
AAAI Conference on Artificial Intelligence (**AAAI**), 2026.
Acceptance rate: 17.6%
 30. *GameChat: Multi-LLM Dialogue for Safe, Agile, and Socially Optimal Multi-Agent Navigation in Constrained Environments.*
Vagul Mahadevan, **Shangdong Zhang**, Rohan Chandra.
IEEE International Symposium on Multi-Robot and Multi-Agent Systems (**MRS**), 2025.
Acceptance rate: 47.8%
 29. *Towards Provable Emergence of In-Context Reinforcement Learning.*
Jiuqi Wang, Rohan Chandra, **Shangdong Zhang.**
Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
Acceptance rate: 24.52%

28. *Finite Sample Analysis of Linear Temporal Difference Learning with Arbitrary Features.*
Zixuan Xie*, Xinyu Liu*, Rohan Chandra, **Shangdong Zhang**.
 Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
 Acceptance rate: 24.52%
27. *Counterfactual Explanations for Continuous Action Reinforcement Learning.*
 Shuyang Dong, **Shangdong Zhang**, Lu Feng.
 International Joint Conference on Artificial Intelligence (**IJCAI**), 2025.
 Acceptance rate: 19.3%
26. *Towards Large Language Models that Benefit for All: Benchmarking Group Fairness in Reward Models.*
 Kefan Song, Jin Yao, Runnan Jiang, Rohan Chandra, **Shangdong Zhang**.
 Reinforcement Learning Conference (**RLC**), 2025.
 Acceptance rate: 39%
25. *Linear Q-Learning Does Not Diverge in L^2 : Convergence Rates to a Bounded Set.*
Xinyu Liu*, Zixuan Xie*, **Shangdong Zhang**.
 International Conference on Machine Learning (**ICML**), 2025.
 Acceptance rate: 26.9%
24. *Transformers Can Learn Temporal Difference Methods for In-Context Reinforcement Learning.*
Jiuqi Wang*, Ethan Blaser*, Hadi Daneshmand, **Shangdong Zhang**.
 International Conference on Learning Representations (**ICLR**), 2025.
 Acceptance rate: 32.08%.
QuantCo Spotlight Award at the ICML Workshop on In-Context Learning, 2024.
23. *Revisiting a Design Choice in Gradient Temporal Difference Learning.*
Xiaochi Qian, **Shangdong Zhang**.
 International Conference on Learning Representations (**ICLR**), 2025.
 Acceptance rate: 32.08%.
22. *Efficient Policy Evaluation with Safety Constraint for Reinforcement Learning.*
Claire Chen*, Shuze Liu*, **Shangdong Zhang**.
 International Conference on Learning Representations (**ICLR**), 2025.
 Acceptance rate: 32.08%.
21. *Doubly Optimal Policy Evaluation for Reinforcement Learning.*
Shuze Liu, Claire Chen, **Shangdong Zhang**.
 International Conference on Learning Representations (**ICLR**), 2025.
 Acceptance rate: 32.08%.
20. *Efficient Multi-Policy Evaluation for Reinforcement Learning.*
Shuze Liu, Claire Chen, **Shangdong Zhang**.
 AAAI Conference on Artificial Intelligence (**AAAI**), 2025.
 Acceptance rate: 23.4%. **Oral presentation (4.6%)**.
19. *Efficient Policy Evaluation with Offline Data Informed Behavior Policy Design.*
Shuze Liu, **Shangdong Zhang**.
 International Conference on Machine Learning (**ICML**), 2024.
 Acceptance rate: 27.5%
18. *On the Convergence of SARSA with Linear Function Approximation.*
Shangdong Zhang, Remi Tachet des Combes, Romain Laroche.
 International Conference on Machine Learning (**ICML**), 2023.
 Acceptance rate: 28%
17. *A Deeper Look at Discounting Mismatch in Actor-Critic Algorithms.*
Shangdong Zhang, Romain Laroche, Harm van Seijen, Shimon Whiteson, Remi Tachet des Combes.
 International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), 2022.
 Acceptance rate: 26%. **Oral presentation.**

16. *Learning Expected Emphatic Traces for Deep RL.*
Ray Jiang, **Shangdong Zhang**, Veronica Chelu, Adam White, Hado van Hasselt.
AAAI Conference on Artificial Intelligence (**AAAI**), 2022.
Acceptance rate: 15%.
15. *Breaking the Deadly Triad with a Target Network.*
Shangdong Zhang, Hengshuai Yao, Shimon Whiteson.
International Conference on Machine Learning (**ICML**), 2021.
Acceptance rate: 21.5%.
14. *Average-Reward Off-Policy Policy Evaluation with Function Approximation.*
Shangdong Zhang*, Yi Wan*, Richard S. Sutton, Shimon Whiteson.
International Conference on Machine Learning (**ICML**), 2021.
Acceptance rate: 21.5%.
13. *Mean-Variance Policy Iteration for Risk-Averse Reinforcement Learning.*
Shangdong Zhang, Bo Liu, Shimon Whiteson.
AAAI Conference on Artificial Intelligence (**AAAI**), 2021.
Acceptance rate: 21.4%.
12. *Learning Retrospective Knowledge with Reverse Reinforcement Learning.*
Shangdong Zhang, Vivek Veeriah, Shimon Whiteson.
Conference on Neural Information Processing Systems (**NeurIPS**), 2020.
Acceptance rate: 20.1%.
11. *GradientDICE: Rethinking Generalized Offline Estimation of Stationary Values.*
Shangdong Zhang, Bo Liu, Shimon Whiteson.
International Conference on Machine Learning (**ICML**), 2020.
Acceptance rate: 21.8%.
10. *Provably Convergent Two-Timescale Off-Policy Actor-Critic with Function Approximation.*
Shangdong Zhang, Bo Liu, Hengshuai Yao, Shimon Whiteson.
International Conference on Machine Learning (**ICML**), 2020.
Acceptance rate: 21.8%.
9. *Deep Residual Reinforcement Learning.*
Shangdong Zhang, Wendelin Boehmer, Shimon Whiteson.
International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), 2020.
Acceptance rate: 23%. **Best Paper Award.**
8. *Mega-Reward: Achieving Human-Level Play without Extrinsic Rewards.*
Yuhang Song, Jianyi Wang, Thomas Lukasiewicz, Zhenghua Xu, **Shangdong Zhang**, Andrzej Wojcicki, Mai Xu.
AAAI Conference on Artificial Intelligence (**AAAI**), 2020.
Acceptance rate: 20.6%.
7. *DAC: The Double Actor-Critic Architecture for Learning Options.*
Shangdong Zhang, Shimon Whiteson.
Conference on Neural Information Processing Systems (**NeurIPS**), 2019.
Acceptance rate: 21.2%.
6. *Generalized Off-Policy Actor-Critic.*
Shangdong Zhang, Wendelin Boehmer, Shimon Whiteson.
Conference on Neural Information Processing Systems (**NeurIPS**), 2019.
Acceptance rate: 21.2%.
5. *Distributional Reinforcement Learning for Efficient Exploration.*
Borislav Mavrin, **Shangdong Zhang**, Hengshuai Yao, Linglong Kong, Kaiwen Wu, Yaoliang Yu.
International Conference on Machine Learning (**ICML**), 2019.
Acceptance rate: 22.6%.

4. *ACE: An Actor Ensemble Algorithm for Continuous Control with Tree Search.*
Shangtong Zhang, Hao Chen, Hengshuai Yao.
 AAAI Conference on Artificial Intelligence (**AAAI**), 2019.
 Acceptance rate: 16.2%. **Spotlight presentation.**
3. *QUOTA: The Quantile Option Architecture for Reinforcement Learning.*
Shangtong Zhang, Borislav Mavrin, Linglong Kong, Bo Liu, Hengshuai Yao.
 AAAI Conference on Artificial Intelligence (**AAAI**), 2019.
 Acceptance rate: 16.2%. **Oral presentation.**
2. *Crossprop: Learning Representations by Stochastic Meta-Gradient Descent in Neural Networks.*
 Vivek Veeriah*, **Shangtong Zhang***, Richard S. Sutton.
 European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD**), 2017.
 Acceptance rate: 27.1%.
1. *A Deep Neural Network for Modeling Music.*
 Pengjing Zhang, Xiaoqing Zheng, Wenqiang Zhang, Siyan Li, Sheng Qian, Wenqi He, **Shangtong Zhang**, Ziyuan Wang.
 International Conference on Multimedia Retrieval (**ICMR**), 2015.
 Acceptance rate: 31%.

Refereed Workshop Papers (Non-Archival)

4. *CRASH: Challenging Reinforcement-Learning Based Adversarial Scenarios For Safety Hardening.*
 Amar Kulkarni, **Shangtong Zhang**, Madhur Behl.
 ML for Autonomous Driving Workshop at AAAI, 2025.
3. *A Deeper Look at Experience Replay.*
Shangtong Zhang, Richard S. Sutton.
 Deep RL Symposium at NIPS, 2017.
2. *Comparing Deep Reinforcement Learning and Evolutionary Methods in Continuous Control.*
Shangtong Zhang, Osmar R. Zaiane.
 Deep RL Symposium at NIPS, 2017.
1. *A Demon Control Architecture with Off-Policy Learning and Flexible Behavior Policy.*
Shangtong Zhang, Richard S. Sutton.
 Hierarchical RL Workshop at NIPS, 2017.

Technical Reports

4. *A Survey of In-Context Reinforcement Learning.*
Amir Moeini, Jiuqi Wang, Jakob Beck, Ethan Blaser, Shimon Whiteson, Rohan Chandra, **Shangtong Zhang**.
 arXiv Preprint, 2025.
3. *Experience Replay Addresses Loss of Plasticity in Continual Learning.*
Jiuqi Wang, Rohan Chandra, **Shangtong Zhang**.
 arXiv Preprint, 2025.
2. *Group Fairness in Multi-Task Reinforcement Learning.*
Kefan Song, Runnan Jiang, Rohan Chandra, **Shangtong Zhang**.
 arXiv Preprint, 2025.
1. *AlphaStar Unplugged: Large Scale Offline Reinforcement Learning.*
 Michael Mathieu*, Sherjil Ozair*, Srivatsan Srinivasan*, Caglar Gulcehre*, **Shangtong Zhang***, Ray Jiang*, Tom Le Paine*, Richard Powell, Konrad Zolna, Julian Schrittwieser, David Choi, Petko Georgiev, Daniel Kenji Toyama, Aja Huang, Roman Ring, Igor Babuschkin, Timo Ewalds, Mahyar Bordbar, Sarah Henderson, Sergio Gomez Colmenarejo, Aaron van den Oord, Wojciech M. Czarnecki,

Nando de Freitas, Oriol Vinyals.
arXiv Preprint, 2023.

GRANTS

13. *Assumption-Carrying Motion Planning for Safe Autonomous Driving.*
CCI Central Virginia Node, **PI**, Total \$100,000, My Share \$70,000 2026 - 2027
12. *Geometry-Calibrated Reinforcement Learning for Cooperative Multi-drone Tracking.*
CCI Central Virginia Node, **Co-PI**, Total \$100,000, My Share \$30,000 2026 - 2027
11. *Learning at Inference: Post-Training LLMs for In-Context Reinforcement Learning.*
Google Awards for Machine Learning Research and Education with TPUs
Sole PI, Total \$50,000 + \$50K cloud credits 2026 - 2027
10. *Towards Closing the Last-Mile Gap in Formal Mathematics.*
4-VA at UVA, **PI**, Total \$30,000, My Share \$30,000 2026 - 2027
9. *Towards Safe and Efficient Automated Vehicle Platooning through Reinforcement Learning from Virtual Reality.*
CCI-CVN-UVA, **PI**, Total \$50,000, My Share \$25,000 2026
8. *Continual Skill Acquisition through In-Context Reinforcement Learning.*
Cisco Research, **Sole PI**, Total \$75,000 2025 - 2026
7. *Reinforced Sensory Intelligence for Safe Autonomous Driving.*
CCI Central Virginia Node, **PI**, Total \$100,000, My Share \$60,000 2025 - 2026
6. *CAREER: Revolutionizing the Evaluation of AI Agents with Online and Offline Data.*
NSF 2442098, **Sole PI**, Total \$600,000, with additional \$10,000 for REU 2025 - 2030
5. *Revolutionizing Domain Generalization with In-Context Reinforcement Learning.*
Nvidia Academic Grant, **Sole PI**, 16,000 A100 Hours (~\$64,000) 2025
4. *Enhancing the Security of Large Language Models Against Persuasion-Based Jailbreak Attacks in Multi-Turn Dialogues.*
CCI Coastal Virginia Node, **Co-PI**, Total \$60,000, My Share \$20,000 2025 - 2026
3. *RAMPART: Reinforcement Against Malicious Penetration by Adversaries in Realistic Topologies.*
DARPA HR001123S0002, **Co-PI** of UVA subaward, Total \$2,150,000, My Share \$77,000 2023 - 2027
2. *SLES: CRASH: Challenging Reinforcement-Learning Based Adversarial Scenarios for Safety Hardening.*
NSF 2331904, **Co-PI**, Total \$800,000, My Share \$400,000 2023 - 2026
1. *III: Small: Moving Offline Learning to Rank Online, from Theory to Practice.*
NSF 2128019, **PI**, Total \$500,000, My Share \$500,000 2021 - 2026

SERVICES

Organizers

CPS Rising Star Workshop 2024, Co-Chair

Grant Panelist and Reviewer

Schmidt Sciences, 2026

National Science Foundation (NSF), 2024, 2026

Austrian Science Fund (FWF), 2025

Virginia's Commonwealth Cyber Initiative (CCI), 2024

Editorial Board

Transaction on Machine Learning Research, Action Editor

2026 - Now

Meta Meta Reviewer

RL Conference 2025, 2026 (Senior Area Chair)

Meta Reviewer

NeurIPS 2025, 2026 (Area Chair)

ICML 2025, 2026 (Area Chair)

ICLR 2024, 2025, 2026, 2027 (Area Chair)

AISTATS 2024, 2025, 2026, 2027 (Area Chair)

L4DC 2025 (Program Committee)

AAMAS 2025 (Senior Program Committee)

RL Conference 2024 (Area Chair)

ACML 2022, 2023, 2024 (Area Chair)

Reviewer

Annals of Applied Probability

NeurIPS 2026 Workshop Proposals

Automatica

Expert Reviewer for TMLR Outstanding Certification 2025

Transactions on Pattern Analysis and Machine Intelligence

Transaction of Machine Learning Research

Journal of Machine Learning Research

Artificial Intelligence Journal

Transactions on Intelligent Systems and Technology

IJCAI 2023

AISTATS 2022

NeurIPS 2020, 2021, 2022, 2023

ICML 2020, 2021, 2022, 2023

AAAI 2020, 2021, 2022, 2023

ICLR 2021, 2022, 2023

SIGCOMM 2022

Offline Reinforcement Learning Workshop at NeurIPS 2020, 2021, 2022

Deep Reinforcement Learning Workshop at NeurIPS 2019, 2020, 2021, 2022

Adaptive and Learning Agents Workshop at AAMAS 2019, 2020

Optimization Foundations for Reinforcement Learning Workshop at NeurIPS 2019

Reinforcement Learning for Real Life Workshop at ICML 2019, 2021

Reinforcement Learning for Real Life Workshop at NeurIPS 2022

Conference Session Chair

AAAI 2023, “Reinforcement Learning Theory & Algorithms”

SUPERVISION

Doctral Students

Ethan Blaser, NSF Graduate Research Fellowship

2023 - Now

Jiuqi Wang

2023 - Now

Xinyu Liu

2024 - Now

Zixuan Xie, UVA Provost’s Fellowship, UVA Engineering Distinguished Fellowship

2024 - Now

Amir Moeini

2024 - Now

Alumni with Theses or Publications

Shuze Liu, PhD → Research Scientist at Meta	2022 - 2025
Kefan Song, Master → PhD at UVA	2023 - 2025
Licheng Luo, Master → PhD at UC Riverside	2023 - 2024
Claire Chen, Undergraduate → Undergraduate at Caltech	2023 - 2025
Vikram Ostrander, Undergraduate (DMP thesis) → Software Engineer at Palantir	2024 - 2025
Vagul Mahadevan, Undergraduate (DMP thesis) → Software Engineer at Metron	2023 - 2025
Steve Zhou, Undergraduate (DMP thesis) → Master at MIT	2023 - 2024
Xiaochi Qian, Undergraduate (Oxford) → Quantitative Researcher at Marshall Wace	2022 - 2024

PhD Dissertation Examination Committees

Zeyu Mu (by Prof. Brian Park)
 Zhaoyuan Su (by Prof. Yue Cheng)
 Yeonbin Son (by Prof. Matthew Bolton)
 Matthew Landers (by Prof. Afsaneh Doryab)
 Shuyang Dong (by Prof. Lu Feng)
 Qianhong Zhao (by Prof. Gang Tao)
 Fengdi Che (U of Alberta, by Prof. Rupam Mahmood and Prof. Dale Schuurmans)
 Shohaib Mahmud (by Prof. Haiying Shen)
 Suraiya Tairin (by Prof. Haiying Shen)
 Tao Jin (by Prof. Farzad Farnoud)
 Fan Yao (by Prof. Hongning Wang and Prof. Haifeng Xu)
 Ingy ElSayed-Aly (by Prof. Lu Feng)
 Sudhir Shenoy (by Prof. Afsaneh Doryab)
 Chuanhao Li (by Prof. Hongning Wang)

INVITED TALKS

A Theory on How LLMs Make Decisions

AMA/IMS Spring Research Conference May 2026

Convergence of Value-Based Reinforcement Learning: Advances and Open Problems

Workshop on Advances in ML Theory, Canada, hosted by Prof. Csaba Szepesvari May 2025

Understanding the Training and Inference of Reinforcement Learning

Tsinghua University, hosted by Prof. Hongning Wang June 2024

On the Cheating of Offline Reinforcement Learning

KAUST Rising Stars in AI Symposium Feb 2024

Offline Reinforcement Learning: Current and Future

AAAI New Faculty Highlight Program Feb 2023

Breaking the Deadly Triad in Off-Policy Reinforcement Learning

Department of Computer Science, University of Virginia Mar 2022

School of Computing Science, Simon Fraser University Feb 2022

Department of Electrical & Computer Engineering, University of Waterloo Feb 2022

School of Informatics, University of Edinburgh Oct 2021

Breaking the Deadly Triad with a Target Network

Microsoft Research Summit Oct 2021

Breaking the Deadly Triad in Reinforcement Learning

RL team, DeepMind, hosted by Dr. Hado van Hasselt Sep 2021

Off-Policy Evaluation

Data Fest 2020, Open Data Science Oct 2020

Off-Policy Evaluation and Control

ByteDance AI Lab, Shanghai Oct 2020

Coding Deep RL Papers

NIPS MLTrain Workshop, Long Beach Dec 2019

Off-Policy Actor-Critic Algorithms

TEACHING & OUTREACH**University of Virginia**

CS 6771: Reinforcement Learning	Spring 2026
CS 4771: Reinforcement Learning	Fall 2025
CS 6316: Machine Learning	Spring 2024, Spring 2025
CS 4501: Reinforcement Learning	Fall 2024
CS 4501: Optimization	Fall 2023
CS 6501: Topics in Reinforcement Learning	Fall 2022

Tutorial: What GenAI Can and Cannot Do?

UVA Innovation Hub at Charlottesville Middle School	July 2025
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Charlottesville High School – Link Lab Mentorship Program

Gravity Battery & Wearable GPS	Fall 2024
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OPEN SOURCE CODE**GitHub Repo: PyTorch Deep RL**

A zoo of popular deep RL algorithms in PyTorch with **3k stars**.

GitHub Repo: Reinforcement Learning: An Introduction

Python implementation of the book *Reinforcement Learning: An Introduction* with **13.8k stars**.

Google Summer of Code (GSoC)

MLPack	2017
The Xapian Project	2014